Quantization Report

1. Constraint

Scenario- ship an English sentiment classifier to laptops with limited download/storage. **Target: checkpoint at least 45% smaller than FP32 with no more than 1.0 pp accuracy loss.** No training was done and no quantization library was used.

2. What I built and tried

I wrote the quantizer in raw PyTorch weights are stored as INT8 with FP32 scales, then rebuilt to FP32 at inference (activations stay FP32). Embeddings, LayerNorm, biases and the final classifier remain FP32. I tested **two mappings** (symmetric, asymmetric) × **three granularities** (per-tensor, per-channel, per-group of 64) = 6 configurations. Accuracy was measured on the 872 SST-2 validation examples latency is the median CPU forward pass.

3. Results

Configuration	Size (MiB)	Saved %	Acc. %	Drop (pp)	RMSE	ms/batch
FP32 baseline	255.45	0.00	91.06	0.00	0	1216.7
Sym per-tensor	132.28	48.22	90.60	0.46	0.00673	1278.3
Sym per-channel	132.44	48.15	90.94	0.11	0.00048	1374.4
Sym per-group (64)	134.84	47.21	91.06	0.00	0.00028	1421.4
Asym per-tensor	132.29	48.21	90.37	0.69	0.00442	1347.0
Asym per-channel (winner)	**132.49**	**48.13**	**91.06**	**0.00**	**0.00041**	**1547.9**
Asym per-group (64)	135.50	46.96	91.06	0.00	0.00024	1443.1

4. What failed or went wrong

- No configuration broke the size/accuracy constraint, but the two per-tensor ones lost the most accuracy (0.46 and 0.69 pp) because one outlier weight stretches the scale for the whole matrix.
- Every INT8 version was slower than FP32 (up to 27% for the winner), since weights must be dequantized on every forward pass and the math is still FP32.
- Asymmetric bug caught: clamping the zero point from a positive-only range like [2, 4] clips large weights. Including real zero in the range before computing the scale fixed it.
- Savings are ~48% not the theoretical 75%, because embeddings are left unquantized.

5. Final decision

I chose INT8 asymmetric per-channel. It gives a 132.49 MiB checkpoint (48.13% smaller), 91.06% accuracy (0.00 pp drop), so it meets both constraints. Symmetric per-group and asymmetric per-group tie on accuracy, but per-channel has a smaller file because it stores far fewer scales. Lower weight RMSE did not always mean better accuracy, so I let the task metric decide.